

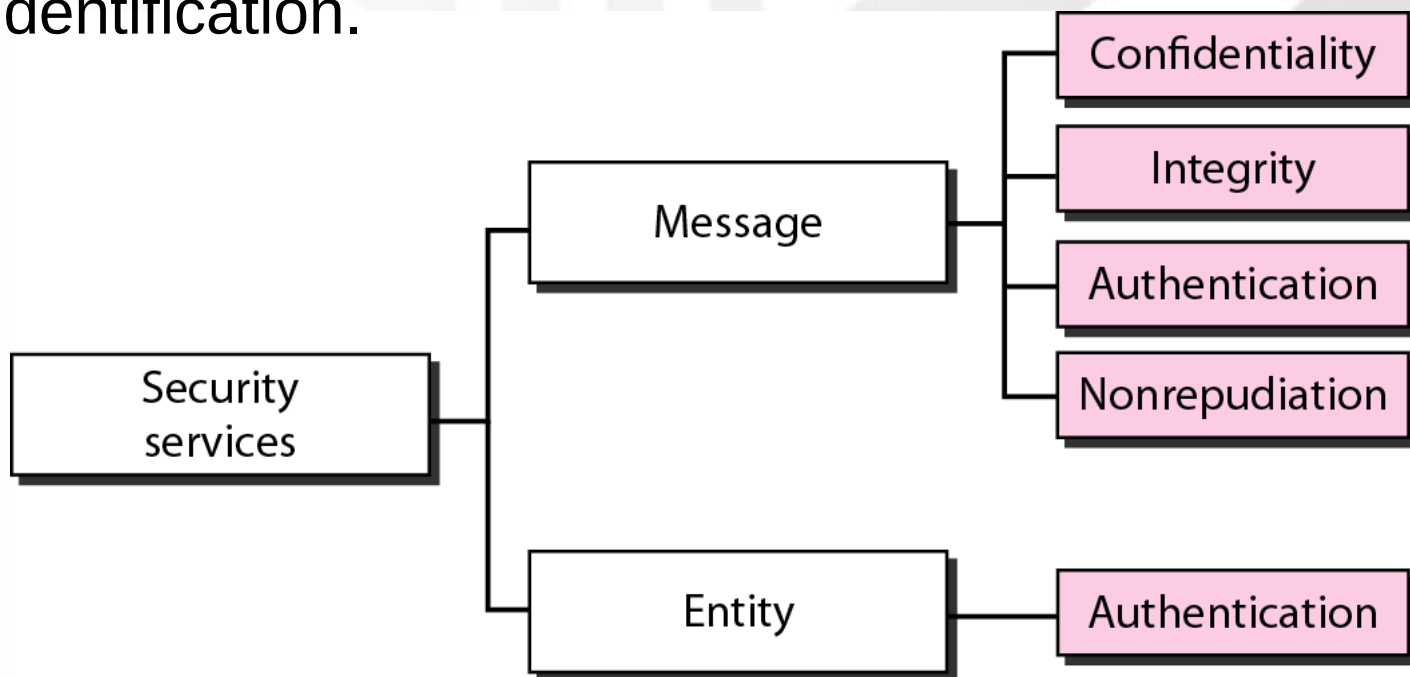
Computer Network and Communication CNS3200

Learning Outcome

- **EXPLAIN HOW DATA IS SECURED IN DATA COMMUNICATION (C4)**
- **EXPLAIN POPULAR APPLICATION PROTOCOLS I.E DNS, SMTP, HTTP, HTTPS (C4)**

SECURITY SERVICES

- Network security can provide five services.
- Four of these services are related to the message exchanged using the network.
- The fifth service provides entity authentication or identification.



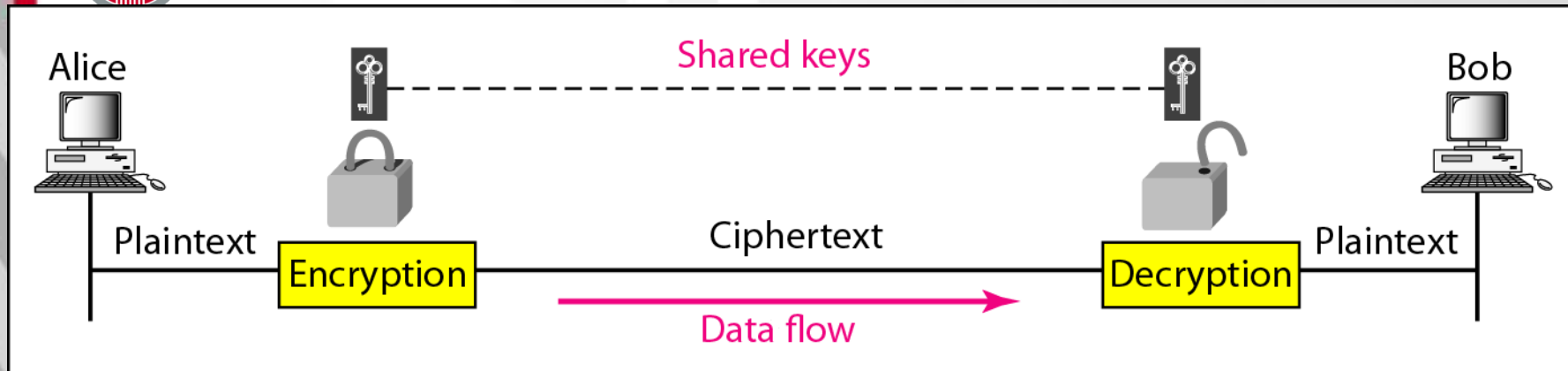
Message Confidentiality



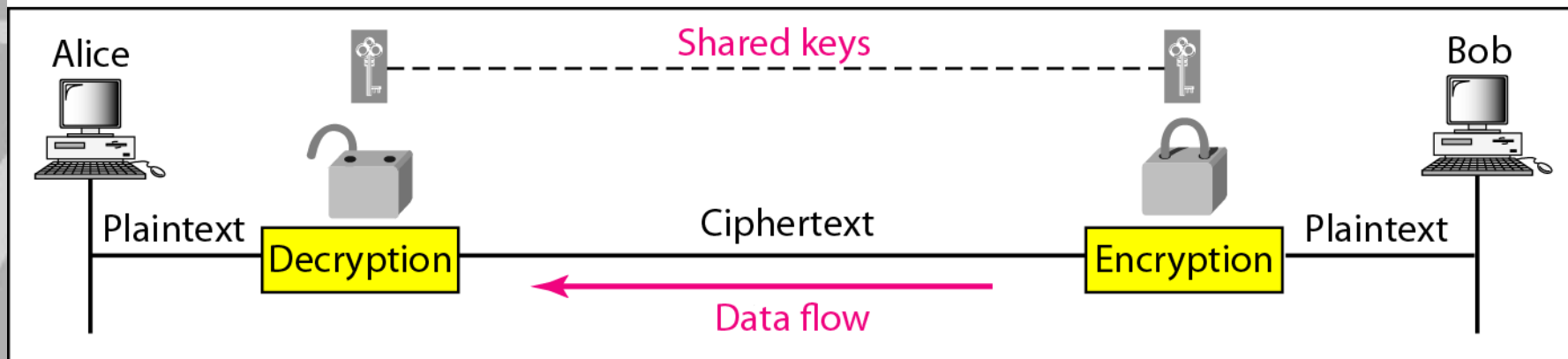
- Message confidentiality or privacy means that the sender and the receiver expect confidentiality.
- The transmitted message must make sense to only the intended receiver.
- To all others, the message must be garbage.
- The message must be encrypted at the sender site and decrypted at the receiver site.
- This can be done using either symmetric-key cryptography or asymmetric-key cryptography.

Confidentiality with Symmetric-Key Cryptography
Confidentiality with Asymmetric-Key Cryptography

Message confidentiality using symmetric keys in two directions

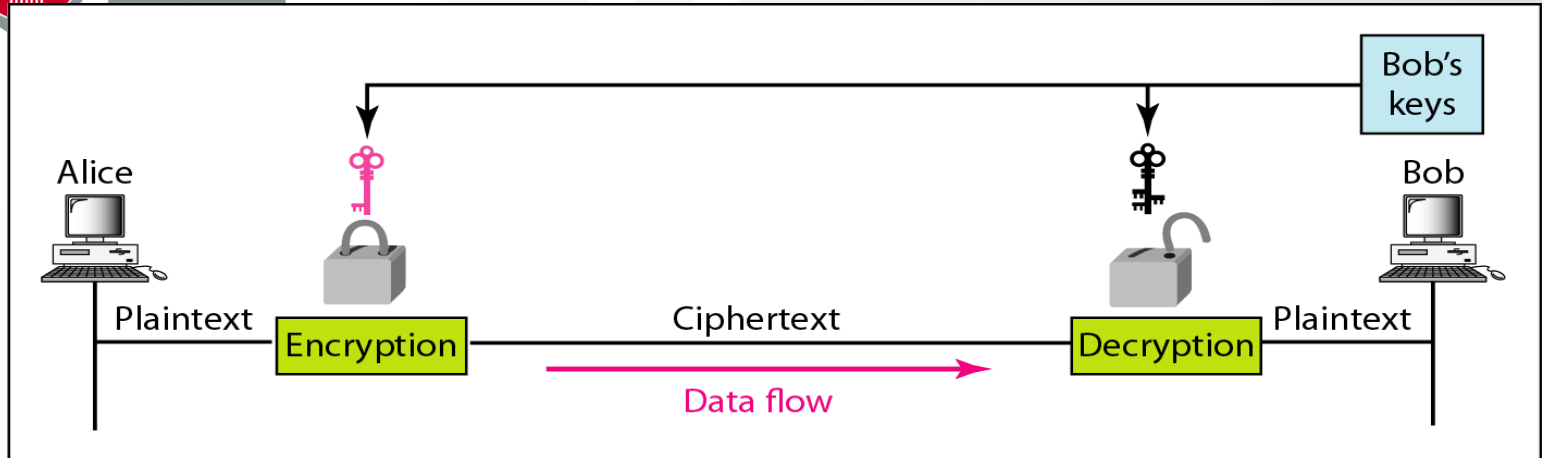


a. A shared secret key can be used in Alice-Bob communication

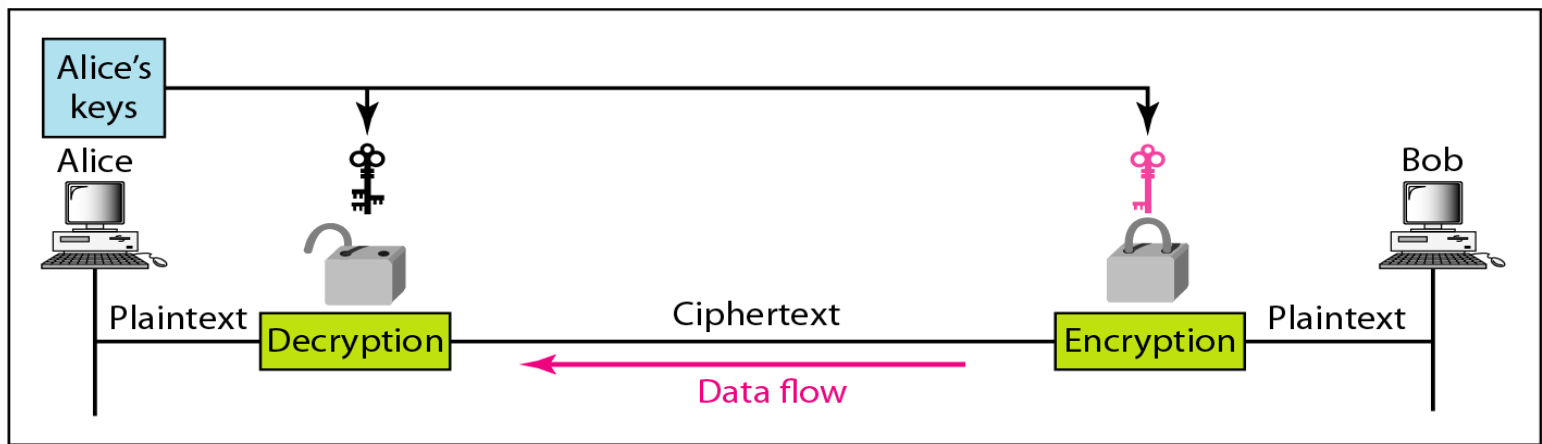


b. A different shared secret key is recommended in Bob-Alice communication

Message confidentiality using asymmetric keys



a. Bob's keys are used in Alice-Bob communication



b. Alice's keys are used in Bob-Alice communication

Message Integrity

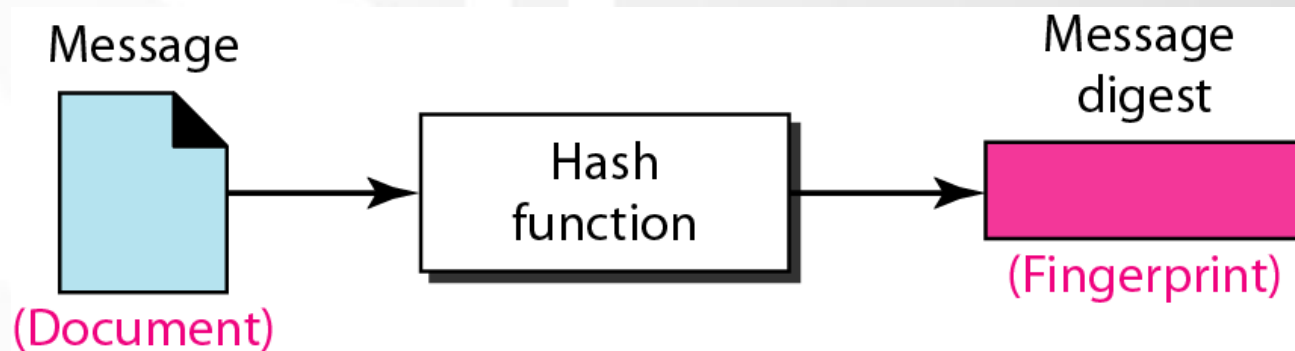
- Message integrity means that the data must arrive at the receiver exactly as they were sent.
- Encryption and decryption provide secrecy, or confidentiality, but not integrity.
- However, on occasion we may not even need secrecy, but instead must have integrity.
 - ✓ Document and Fingerprint
 - ✓ Message and Message Digest
 - ✓ Creating and Checking the Digest
 - ✓ Hash Function Criteria
 - ✓ Hash Algorithms: SHA-1

Note

To preserve the integrity of a document,
both the document and the fingerprint are needed.



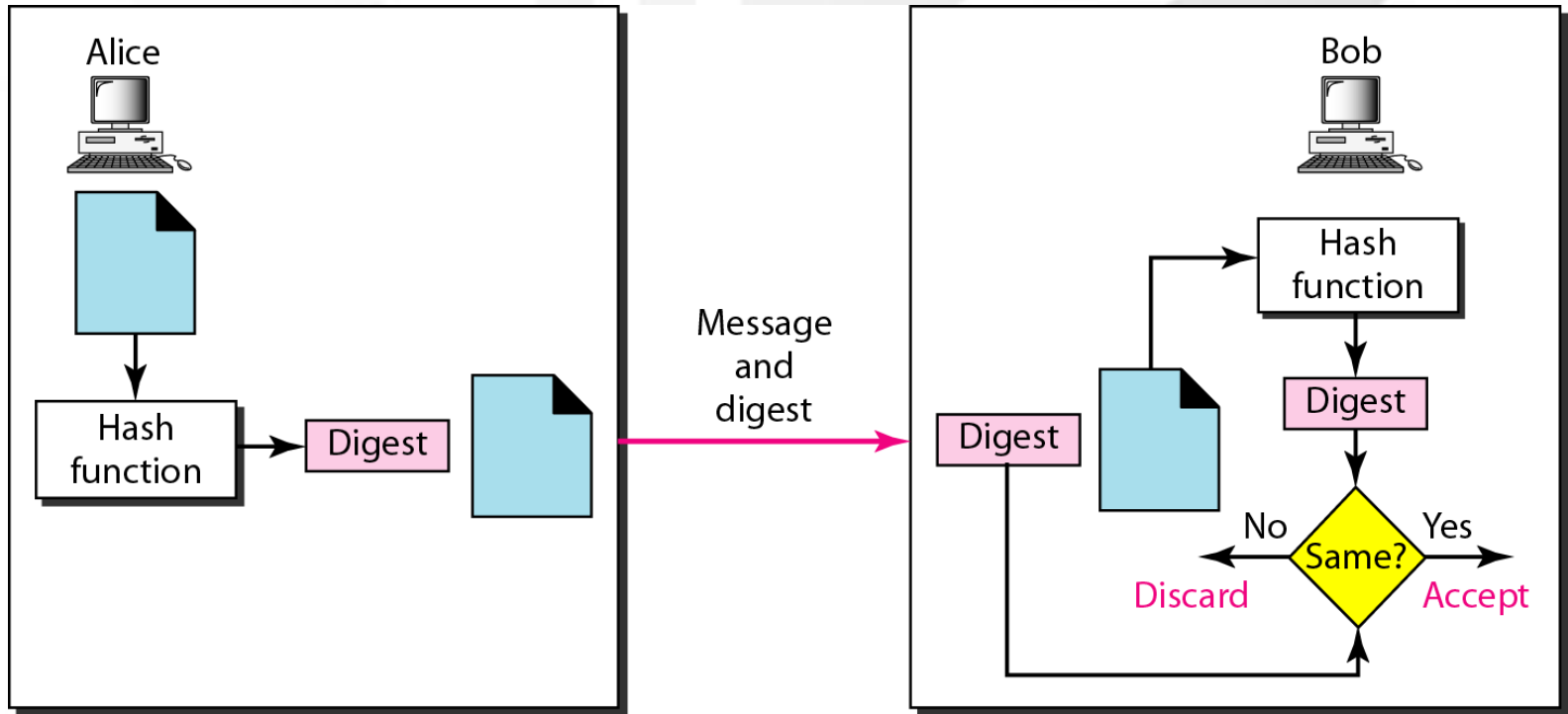
Message and message digest



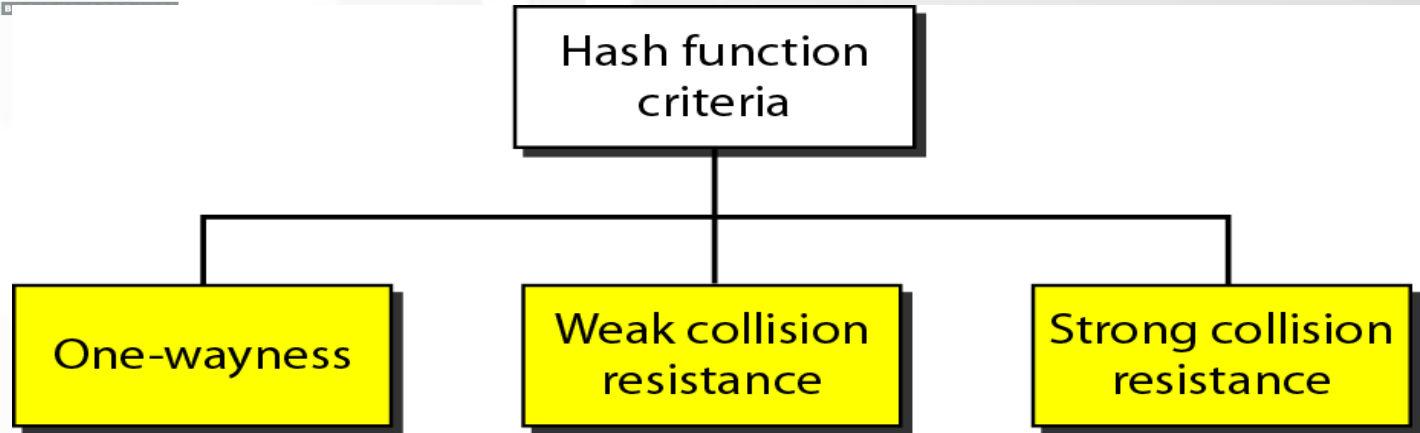
Note

The message digest needs to be kept secret.

Checking integrity



Criteria of a hash function

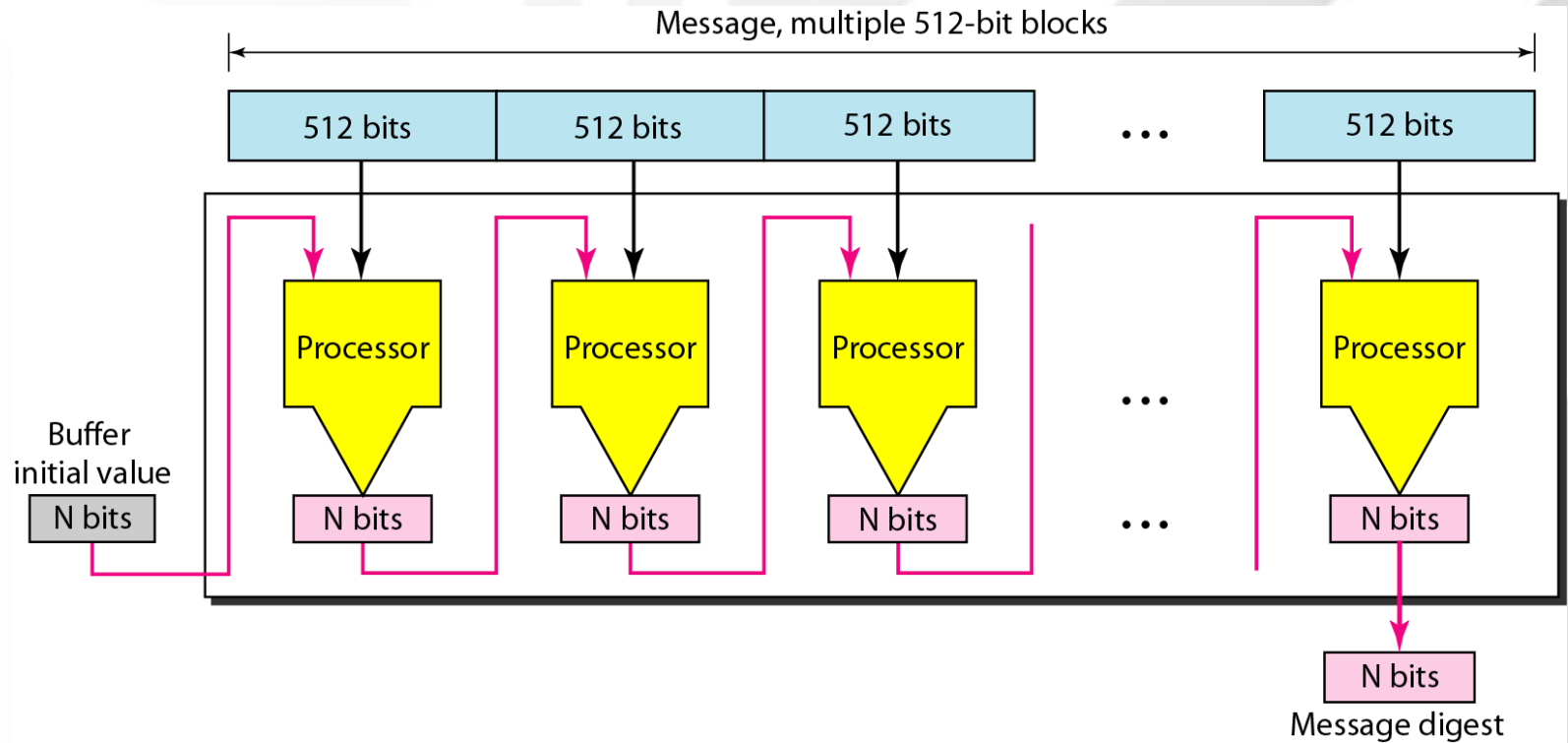


One wayness : Ensure a message digest is created by a one-way hashing function and must not be able to recreate the message from the digest

Weak collision resistance : Ensure that a message cannot be easily be forged

Strong collision resistance: Ensure that we cannot find two messages that hash to the same digest.

Message digest creation

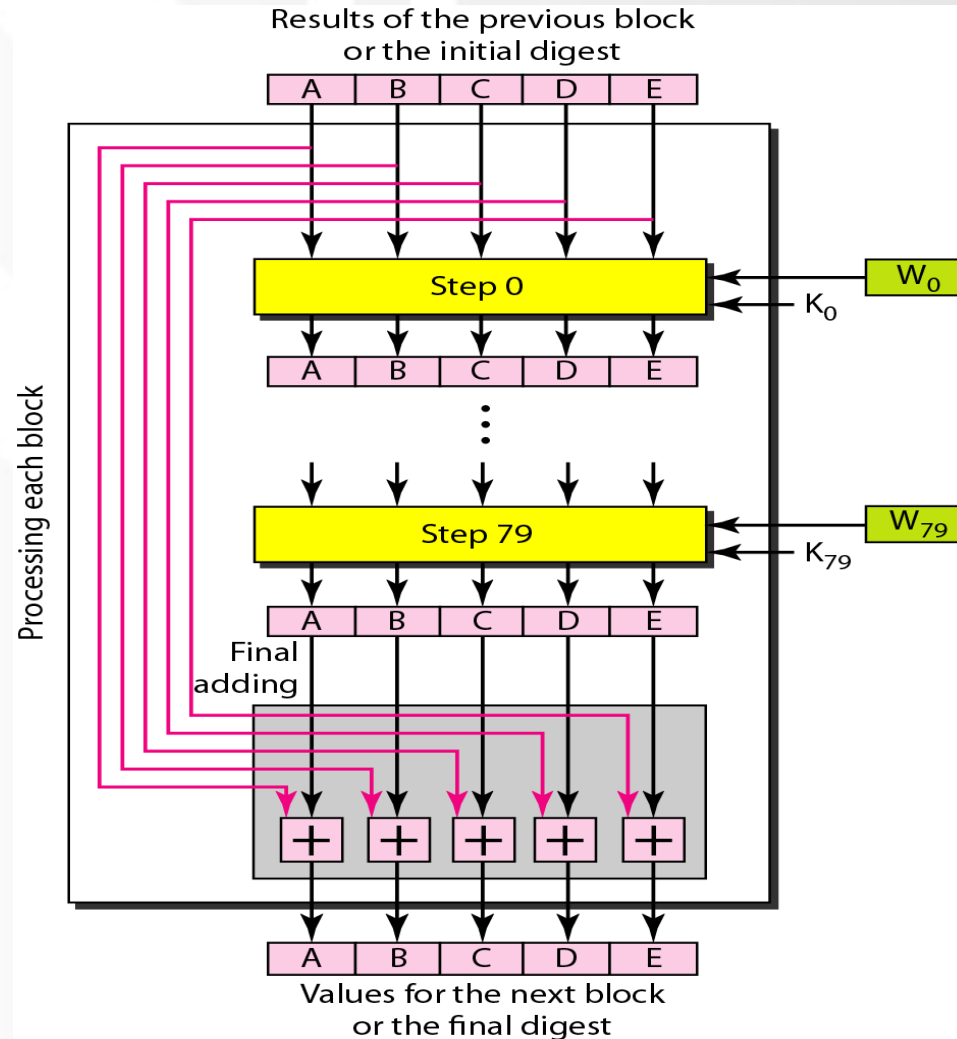


Note

SHA-1 hash algorithms create an N-bit message digest out of a message of 512-bit blocks.

SHA-1 has a message digest of 160 bits (5 words of 32 bits).

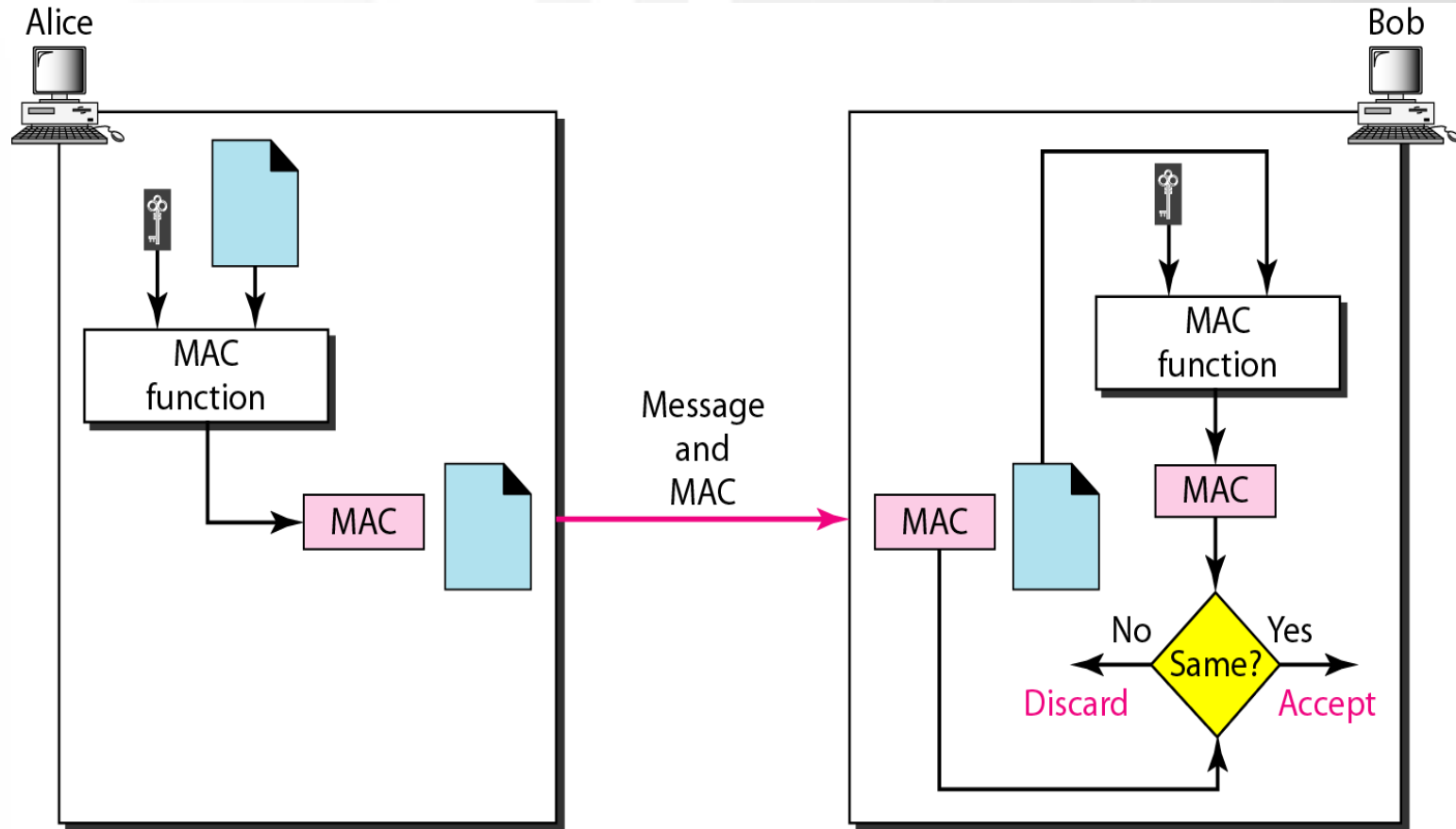
Processing of one block in SHA-1

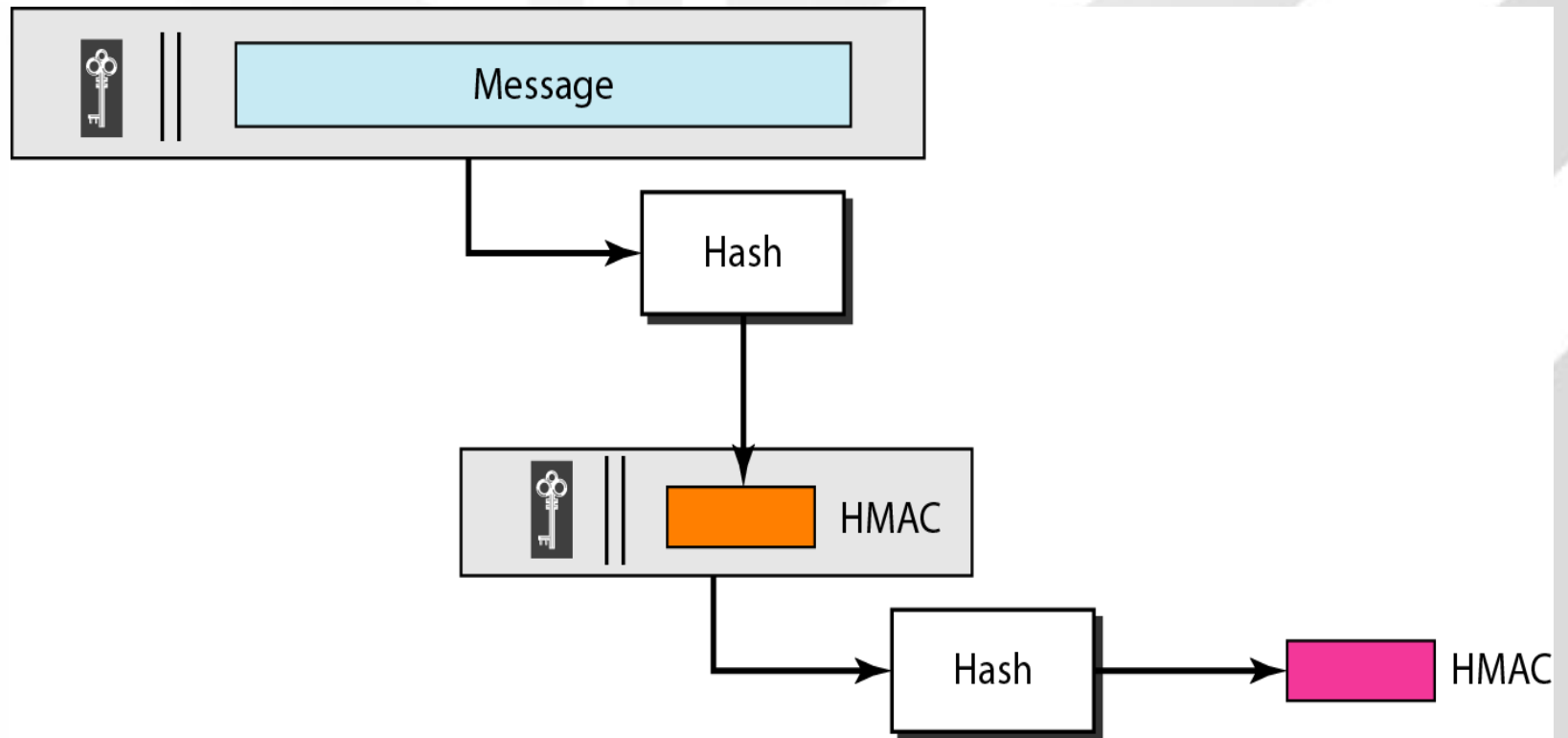


Message Authentication

- Message authentication is a service beyond message integrity.
- In message authentication the receiver needs to be sure of the sender's identity and that an imposter has not sent the message.
- A hash function per se cannot provide authentication.
- The digest created by a hash function can detect any modification in the message, but not authentication.
 - ✓ MAC: A keyed Message Authentication Code
 - ✓ HMAC: Hashed MAC uses keyless hash function such as SHA-1

MAC, created by Alice and checked by Bob







Message Non-repudiation

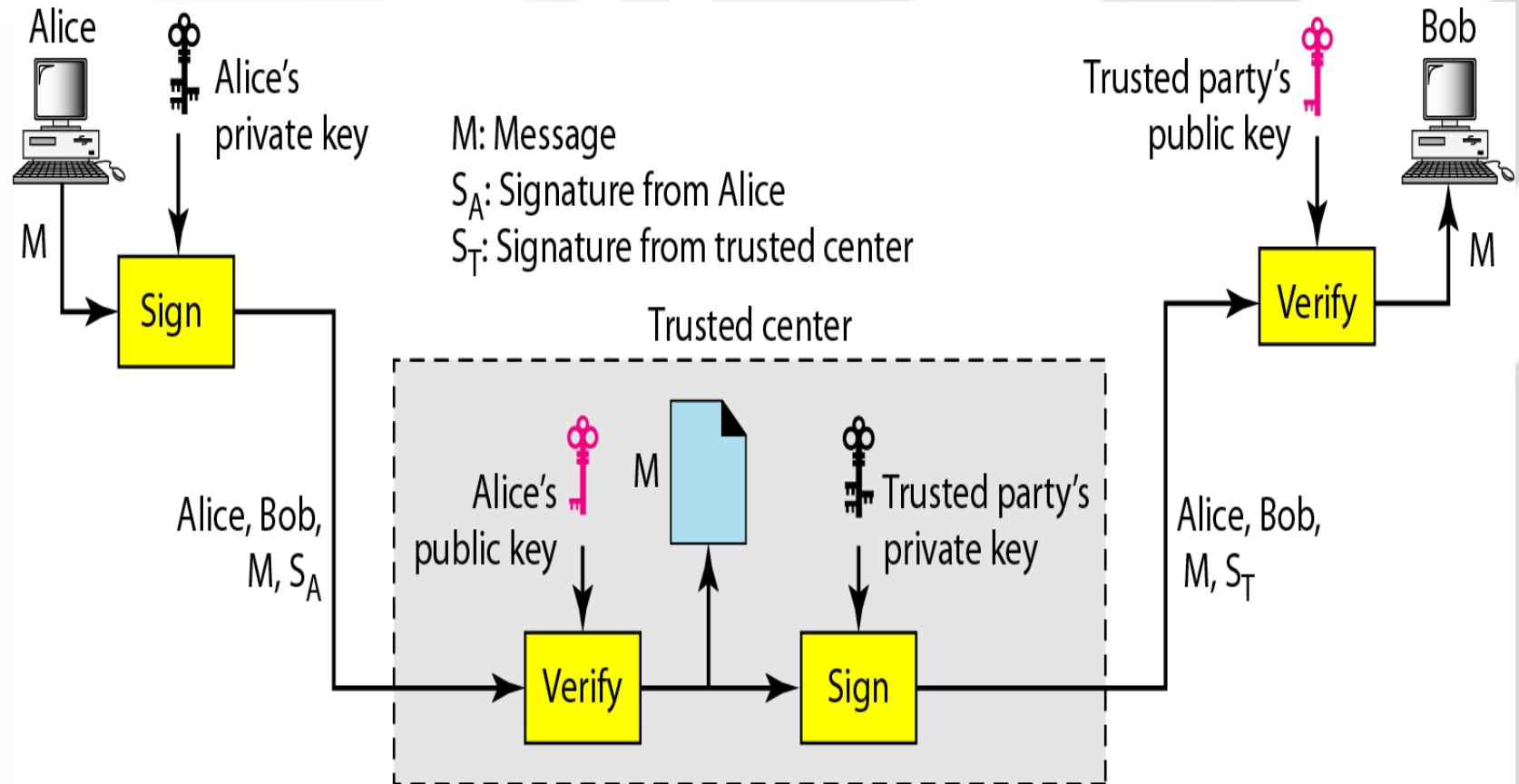
- Message non-repudiation means that a sender must not be able to deny sending a message that he or she, in fact, did send.
- The burden of proof falls on the receiver.

Note

Non-repudiation can be provided using a trusted party.



Using a trusted center for non-repudiation



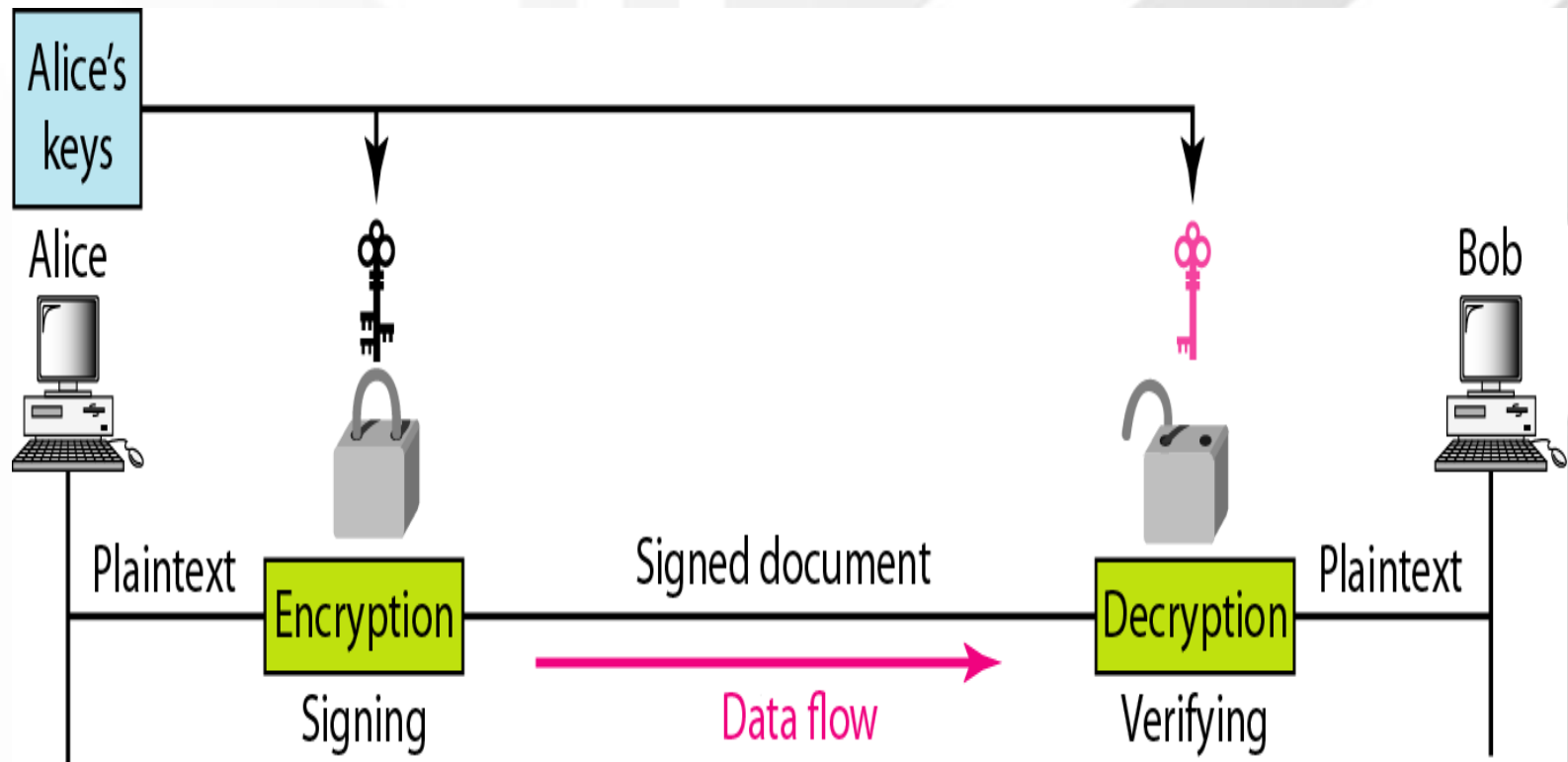
Digital Signature

- When Alice sends a message to Bob, Bob needs to check the authenticity of the sender; he needs to be sure that the message comes from Alice and not Eve.
- Bob can ask Alice to sign the message electronically.
- In other words, an electronic signature can prove the authenticity of Alice as the sender of the message.
- We refer to this type of signature as a digital signature.

Note

A digital signature needs a public-key system.

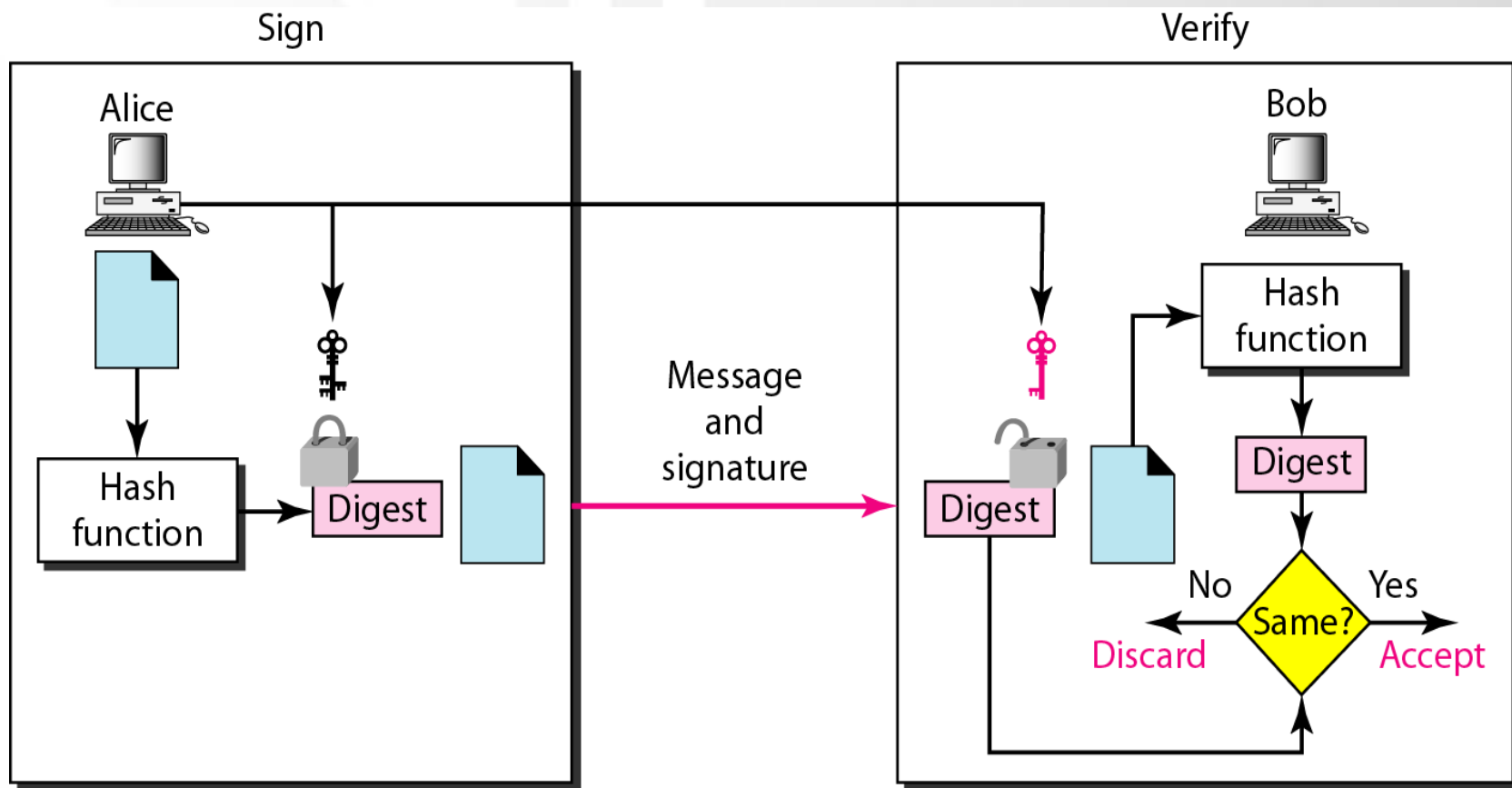
Signing the message itself in digital signature



Note

In a cryptosystem, we use the private and public keys of the receiver;
in digital signature, we use the private and public keys of the sender.

Signing the digest in a digital signature



Note

A digital signature today provides message integrity.
Digital signature provides message authentication.

Entity Authentication

- Entity authentication is a technique designed to let one party prove the identity of another party.
- An entity can be a person, a process, a client, or a server.
- The entity whose identity needs to be proved is called the claimant; the party that tries to prove the identity of the claimant is called the verifier.

Entity Authentication

- Passwords Authentication : **fixed password** and **one-time password**
- Challenge Response Authentication:
 1. Using a symmetric-key cipher
 2. Using a Keyed hash function
 3. Using an symmetric –key cipher
 4. Using digital signature

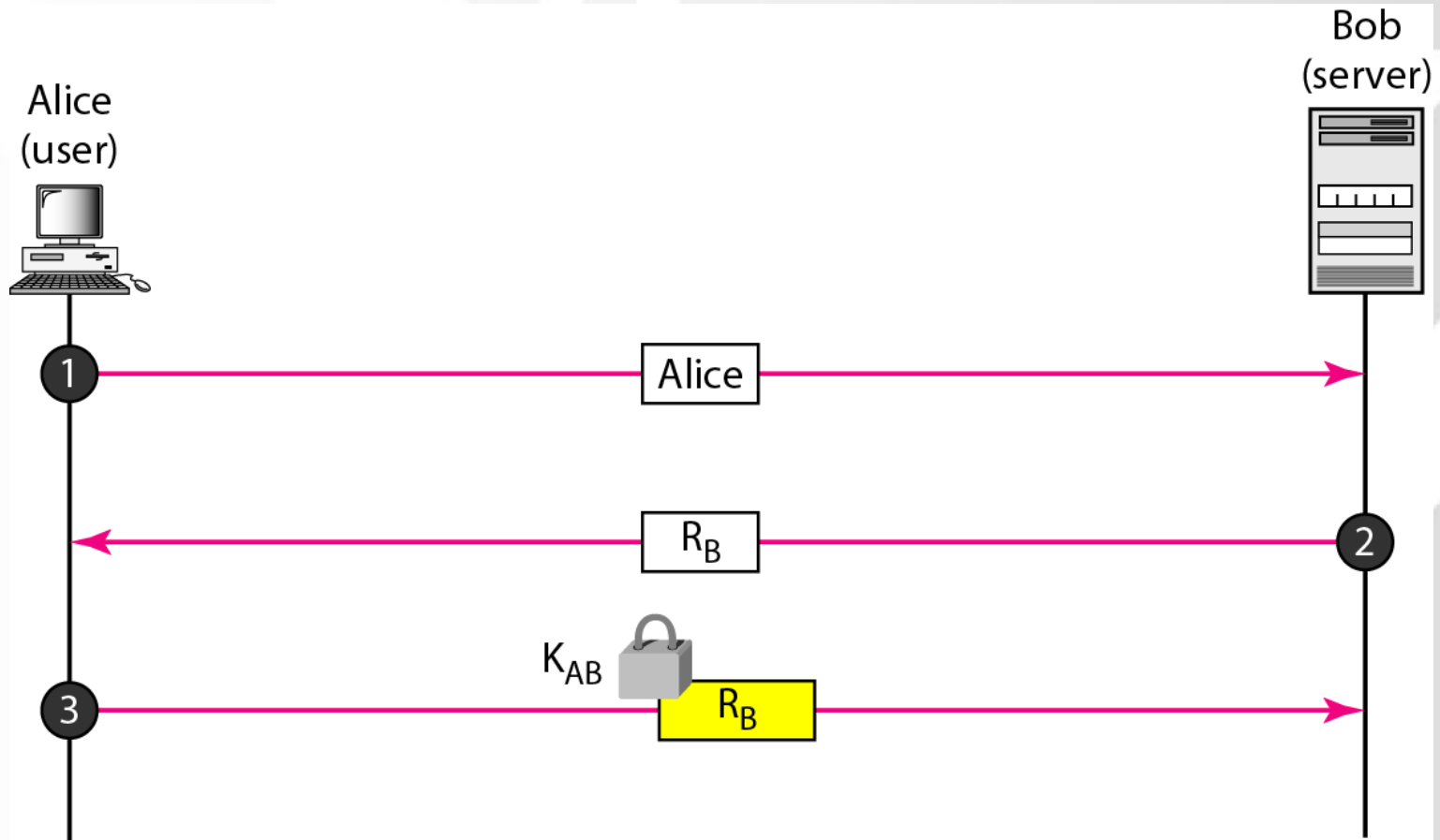
Note

In challenge-response authentication,
the claimant proves that she knows a secret without
revealing it.

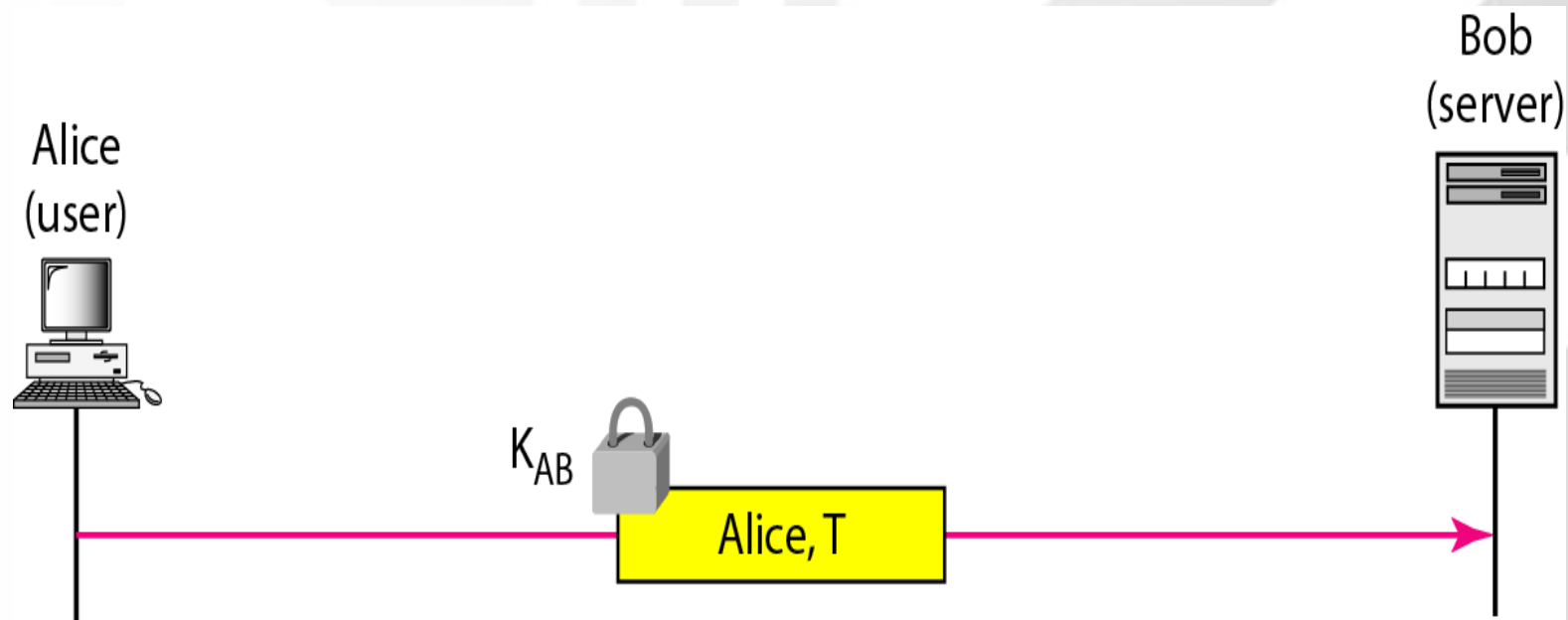
Note

The challenge is a time-varying value sent by the verifier;
the response is the result of a function applied on the
challenge.

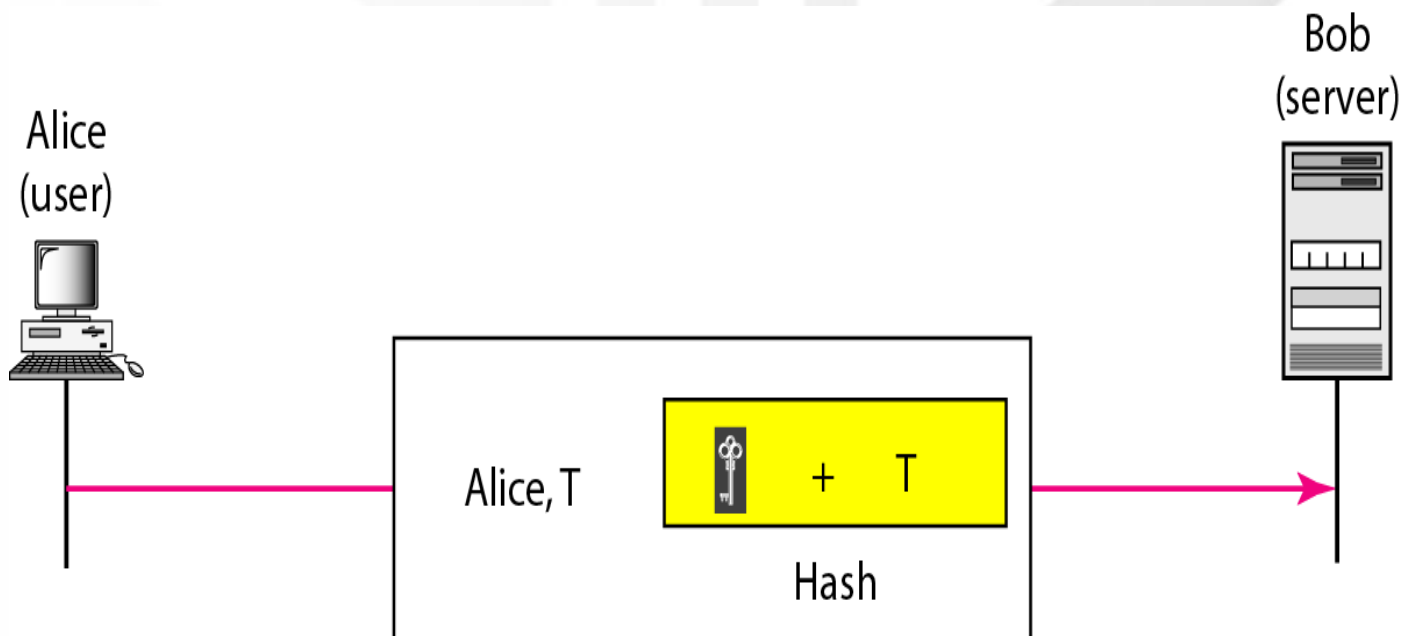
Challenge/response authentication using a nonce



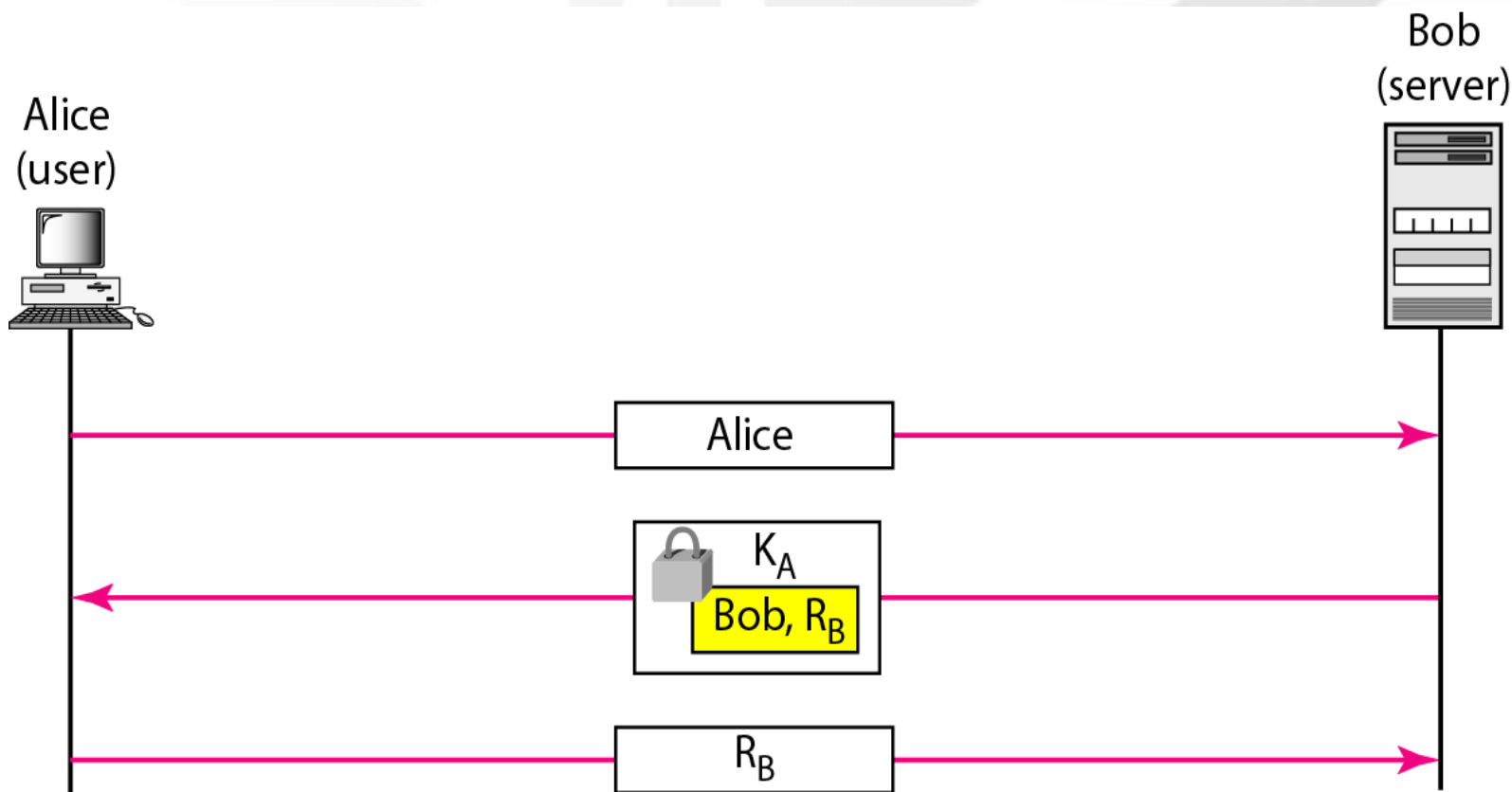
Challenge-response authentication using a timestamp



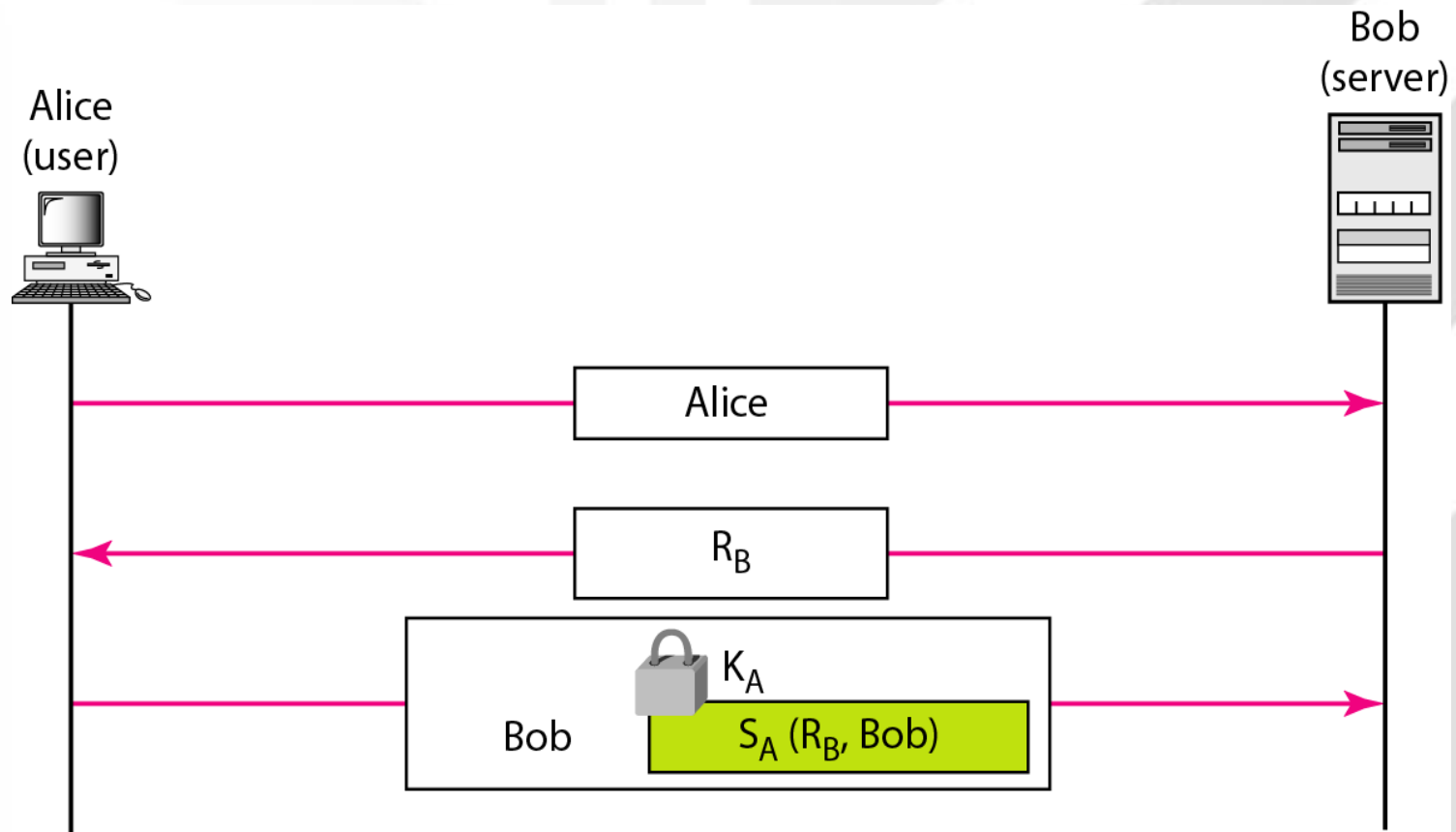
Challenge-response authentication using a keyed-hash function



Authentication, asymmetric-key



Authentication, using digital signature

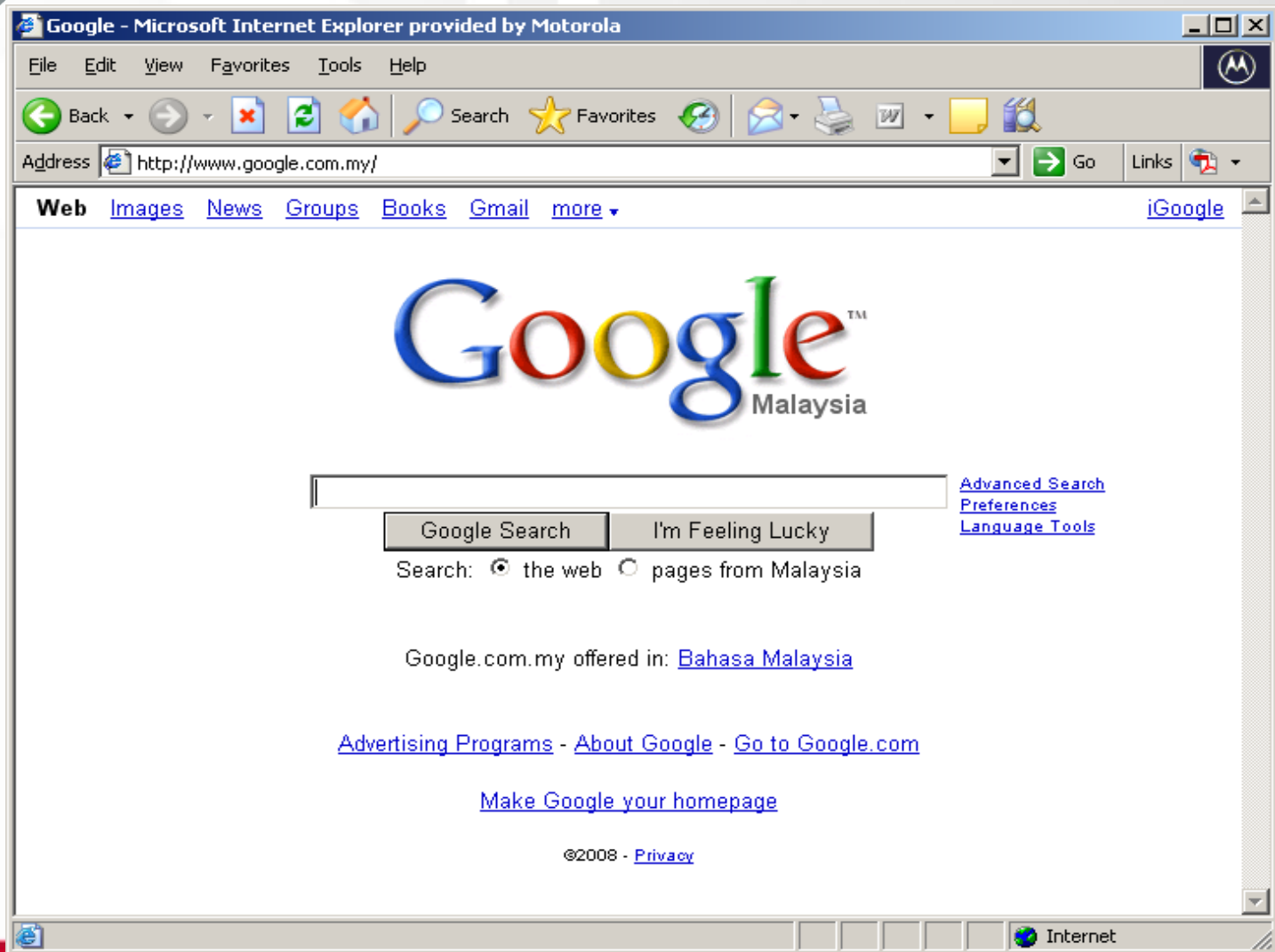




HTTP

*The **WWW** today is a distributed client/server service, in which a client using a browser can access a service using a server. However, the service provided is distributed over many locations called sites.*

Web application





SMTP

One of the most popular Internet services is electronic mail (e-mail). The designers of the Internet probably never imagined the popularity of this application program..



Email

Inbox - Microsoft Outlook

File Edit View Go Tools Actions Help Adobe PDF

E-mail Accounts

You can select an account and change its settings or remove it.

Outlook processes e-mail for these accounts in the following order:

Name	Type
fsktm.upm.edu.my	POP/SMTP (Default)

Buttons: Change... Add... Remove Set as Default Move Up Move Down

Deliver new e-mail to the following location:
Personal Folders [v] New Outlook Data File...

< Back Finish Cancel

WASET ONLINE CONSULTATION FORM: Conferences

set [info@waset.org]
idawaty@fsktm.upm.edu.my

r Idawaty Ahmad,

will get the notification soon.

--- Original Message ---
rom: idawaty@fsktm.upm.edu.my
o: info@waset.org
ent: Friday, July 04, 2008 8:38 AM
ubject: WASET ONLINE CONSULTATION FORM: Conferences

WASET ONLINE CONSULTATION FORM

Date : 04/07/2008 01:38:45 IP Adress :119.40.118.201

Full Name	Idawaty Ahmad
Institution	University Putra Malaysia
Country	Malaysia

[Fwd: Fw: PRICE CONTROL]
Fauzan Othman-CMO018C 6/13/2008
FW: Tips for filling petrol
zan 6/12/2008
direktori 12.15pm



Remote Logging -Telnet

It would be impossible to write a specific client/server program for each demand. The better solution is a general-purpose client/server program that lets a user access any application program on a remote computer

```
C:\ Telnet fsktm.upm.edu.my
login: idawaty
Password:
Login incorrect
login: idawaty
Password:
Sun Microsystems Inc.      SunOS 5.10      Generic January 2005
You have new mail.
mfsktm% ls
INBOX.Drafts              INBOX.Trash              nmap-4.62-setup.zip
INBOX.Sent                 mail                     public_html
mfsktm% cd public_html
mfsktm% p^Hls
p^Hls: Command not found
mfsktm% ls
index.html  index1.php  personal  wlect
mfsktm%
```



Domain Name System (DNS)

- ☐ *The names assigned to machines must be carefully selected from a name space with complete control over the binding between the names and IP addresses*
- ☐ *To have a hierarchical name space, a domain name space was designed. In this design the names are defined in an inverted-tree structure with the root at the top. The tree can have only 128 levels: level 0 (root) to level 127*

DNS

□ *DNS is a protocol that can be used in different platforms. In the Internet, the domain name space (tree) is divided into three different sections: generic domains, country domains, and the inverse domain*

Figure 25.1 *Example of using the DNS service*

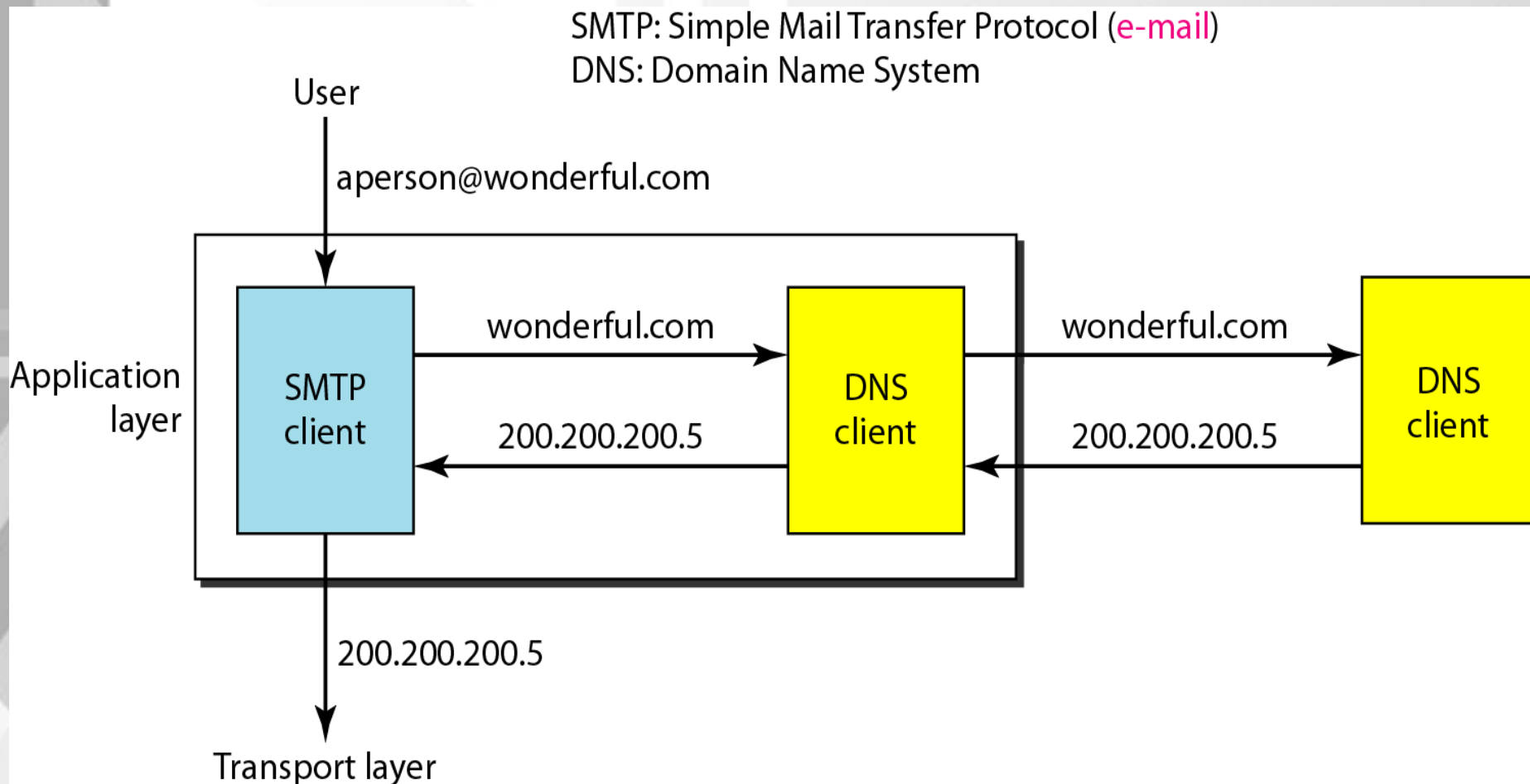


Figure 25.2 *Domain name space*

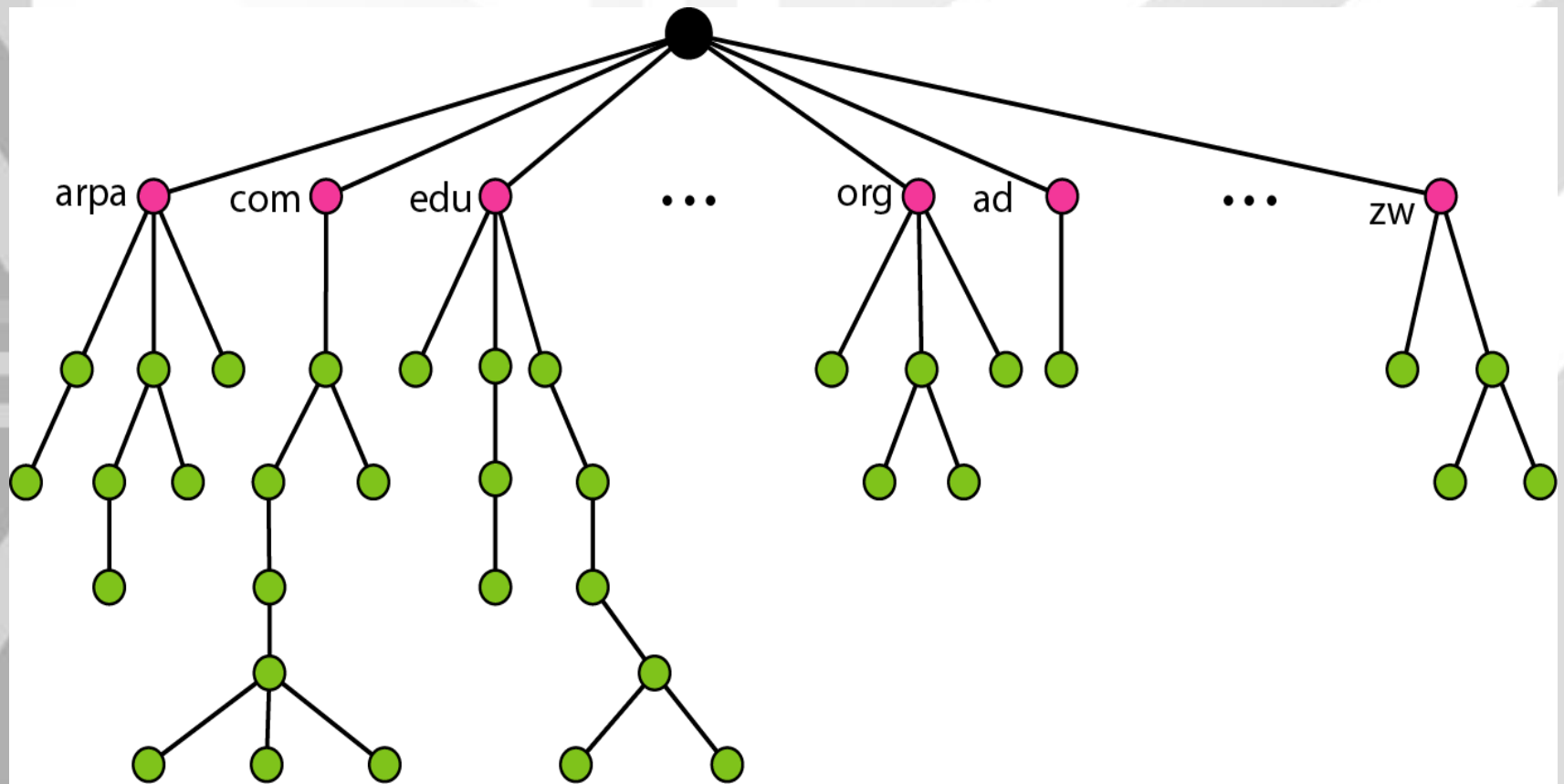


Figure 25.3 *Domain names and labels*

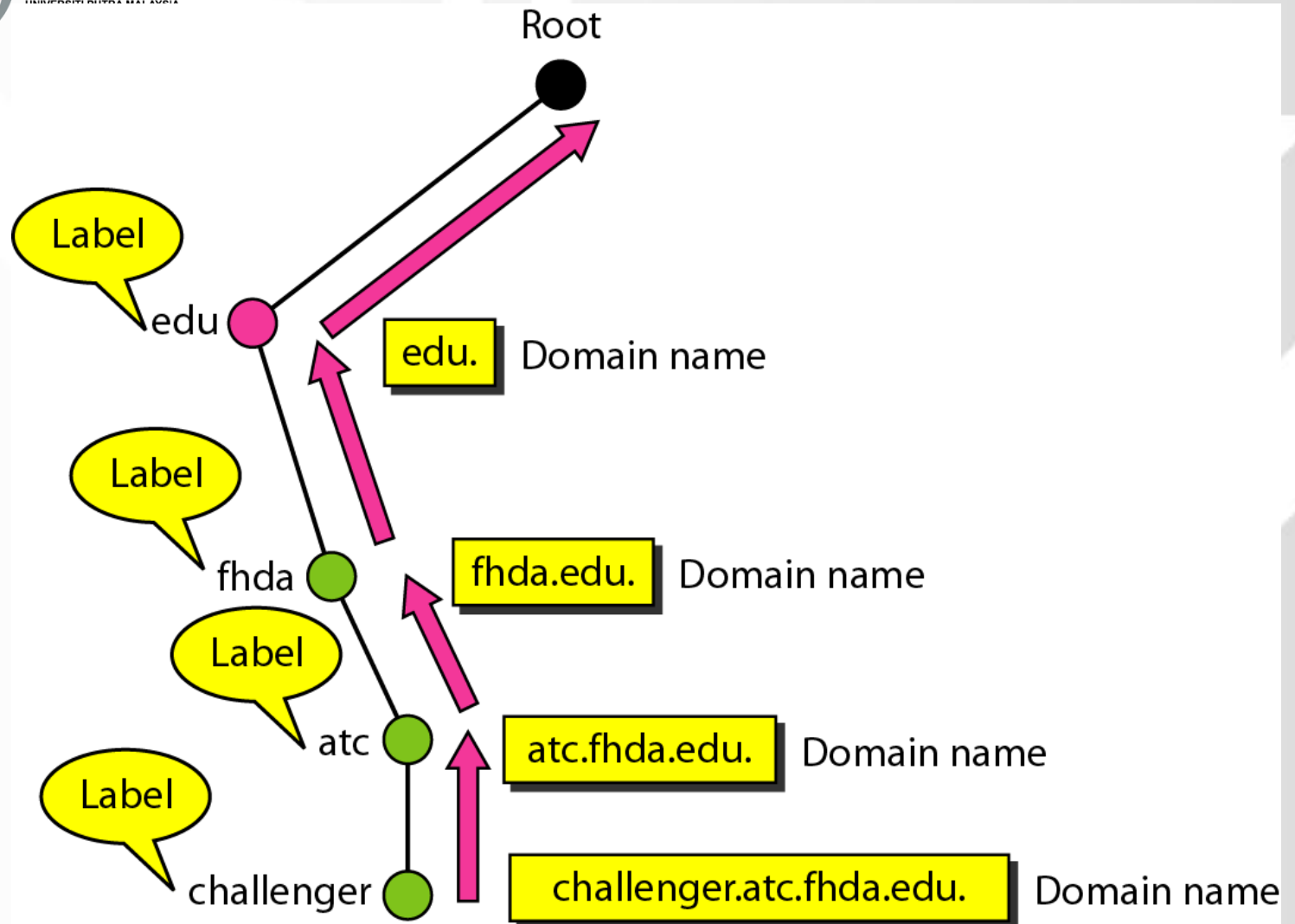




Figure 25.5 *Domains*

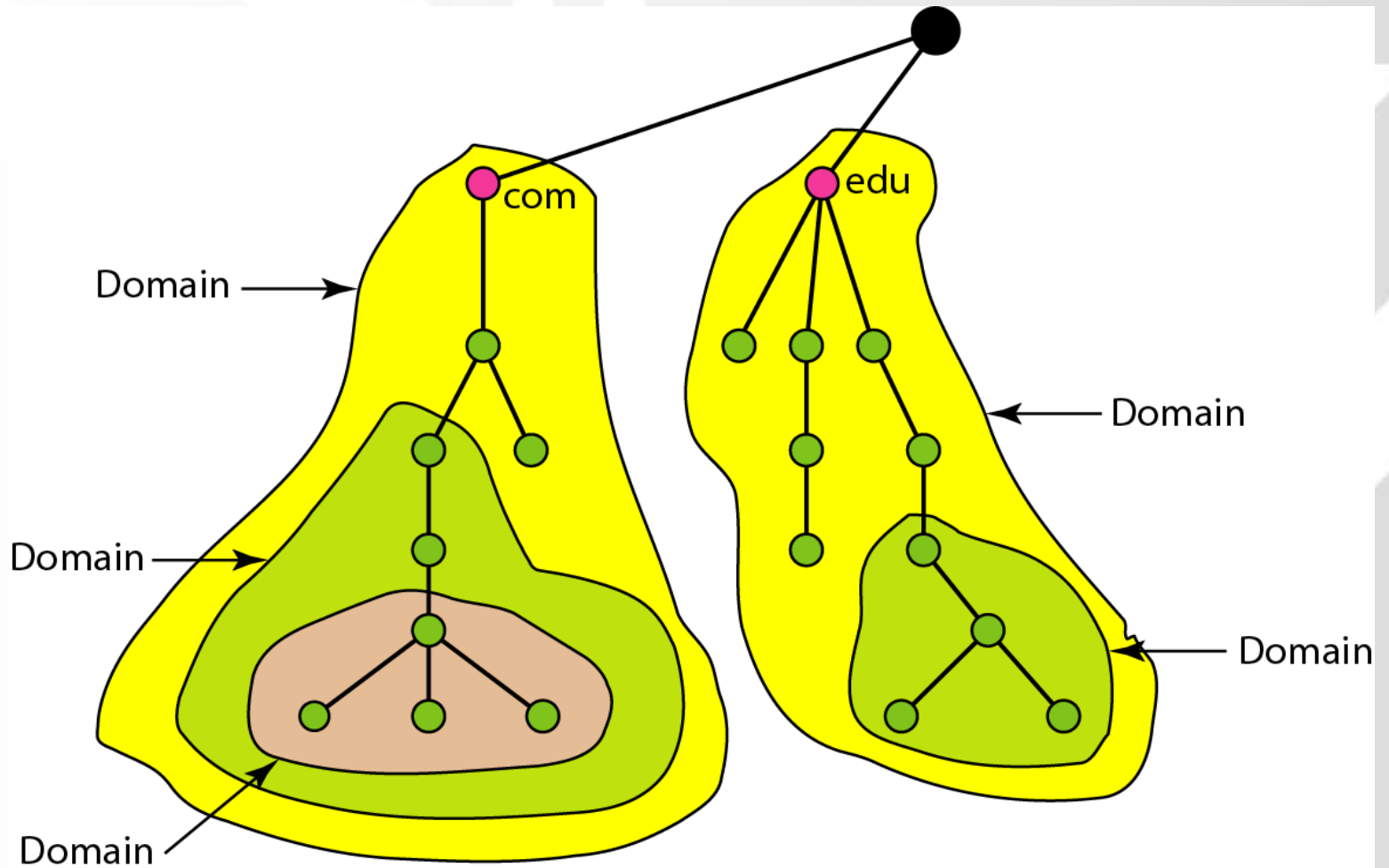


Figure 25.6 *Hierarchy of name servers*

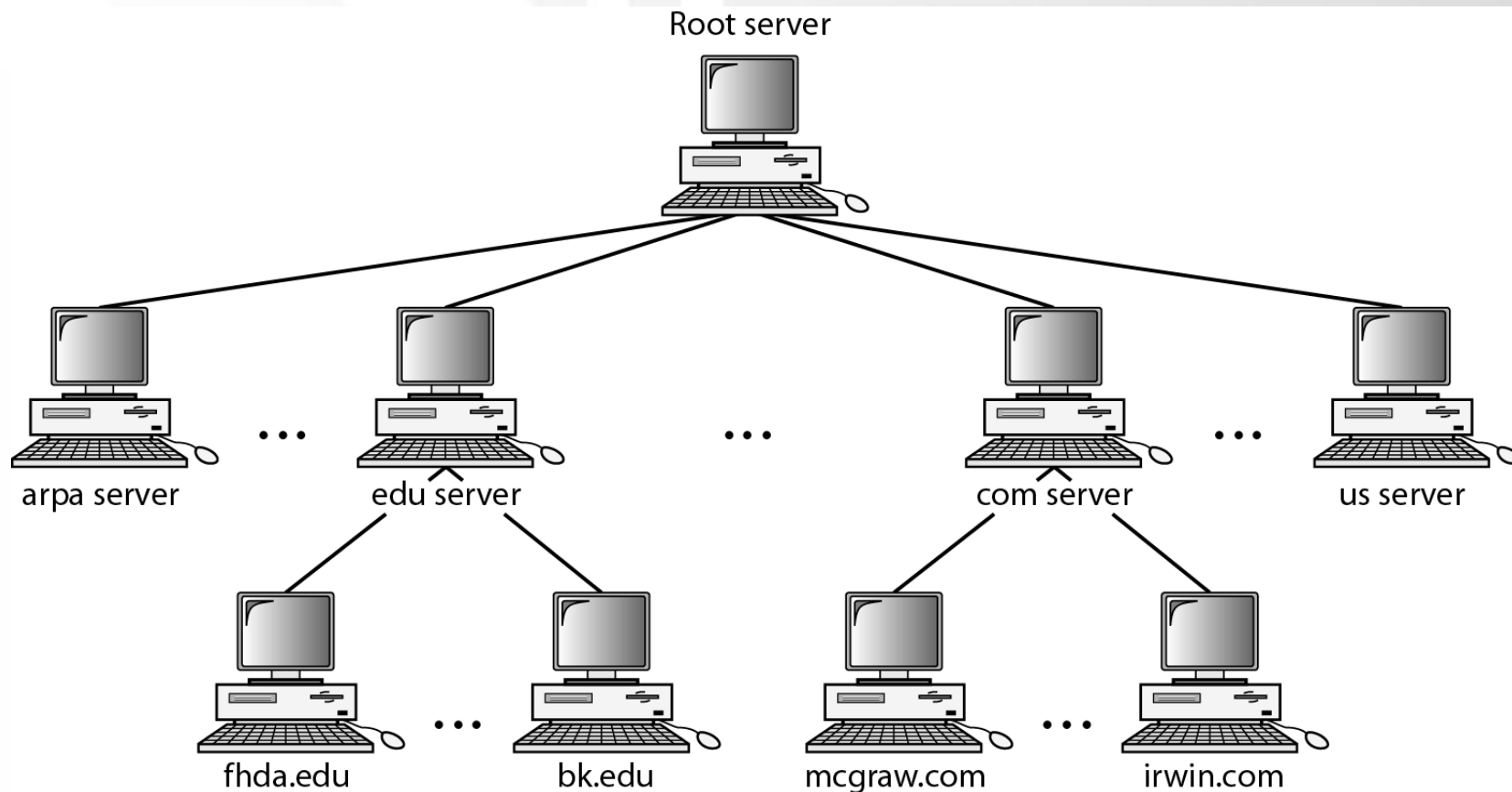
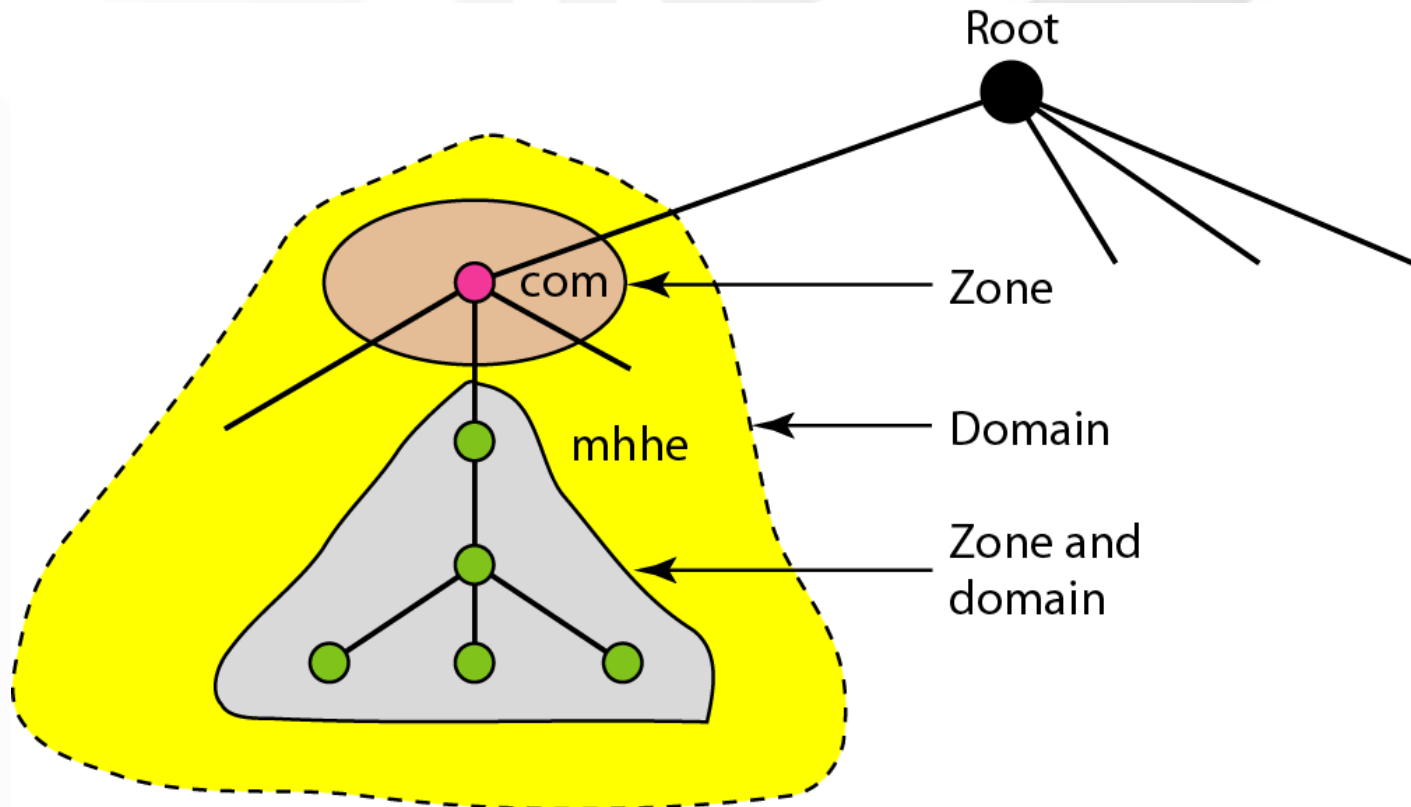


Figure 25.7 *Zones and domains*





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Note

A primary server loads all information from the disk file; the secondary server loads all information from the primary server.

When the secondary downloads information from the primary, it is called zone transfer.

Figure 25.8 *DNS IN THE INTERNET*

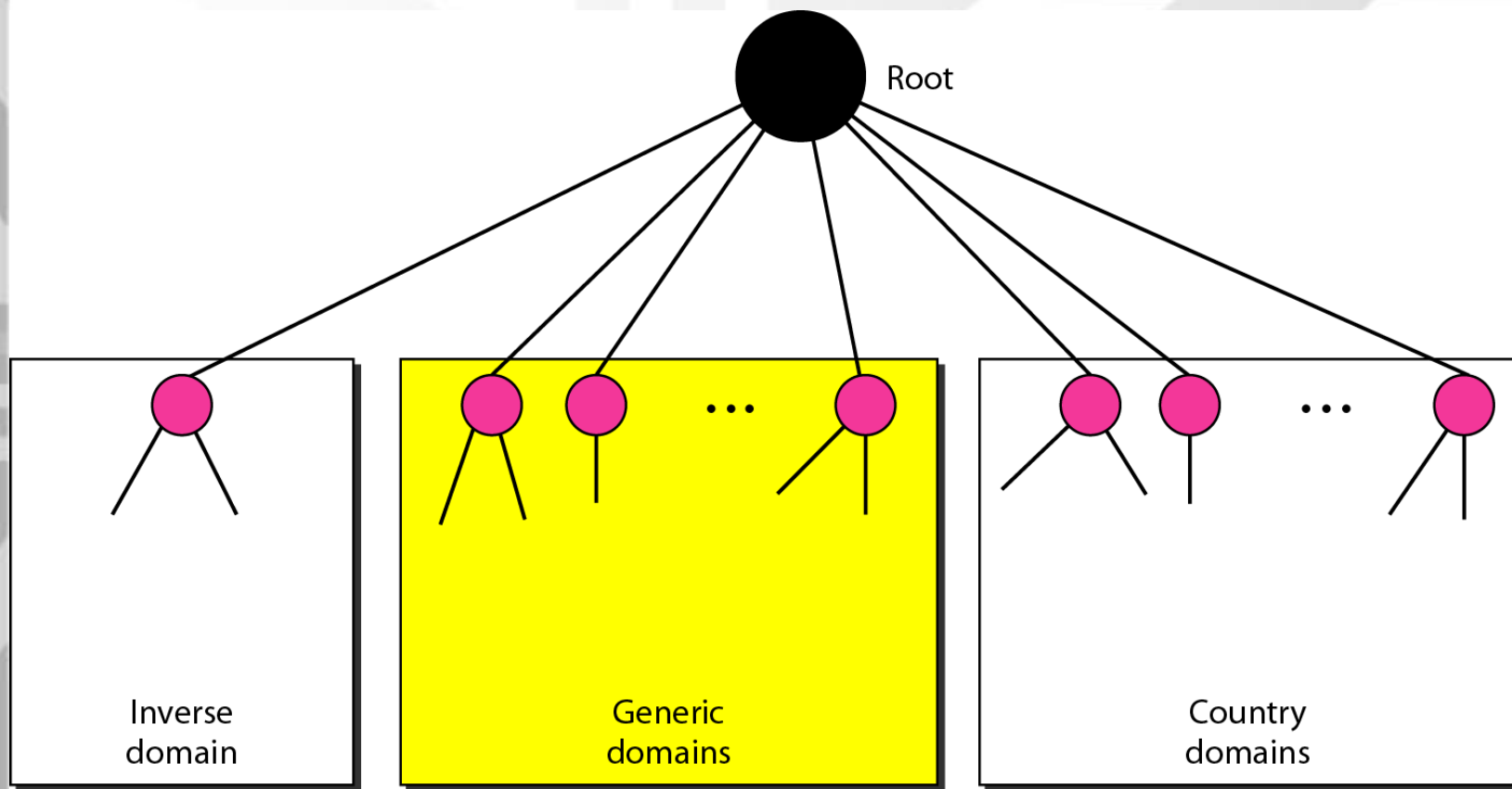


Figure 25.9 *Generic domains*

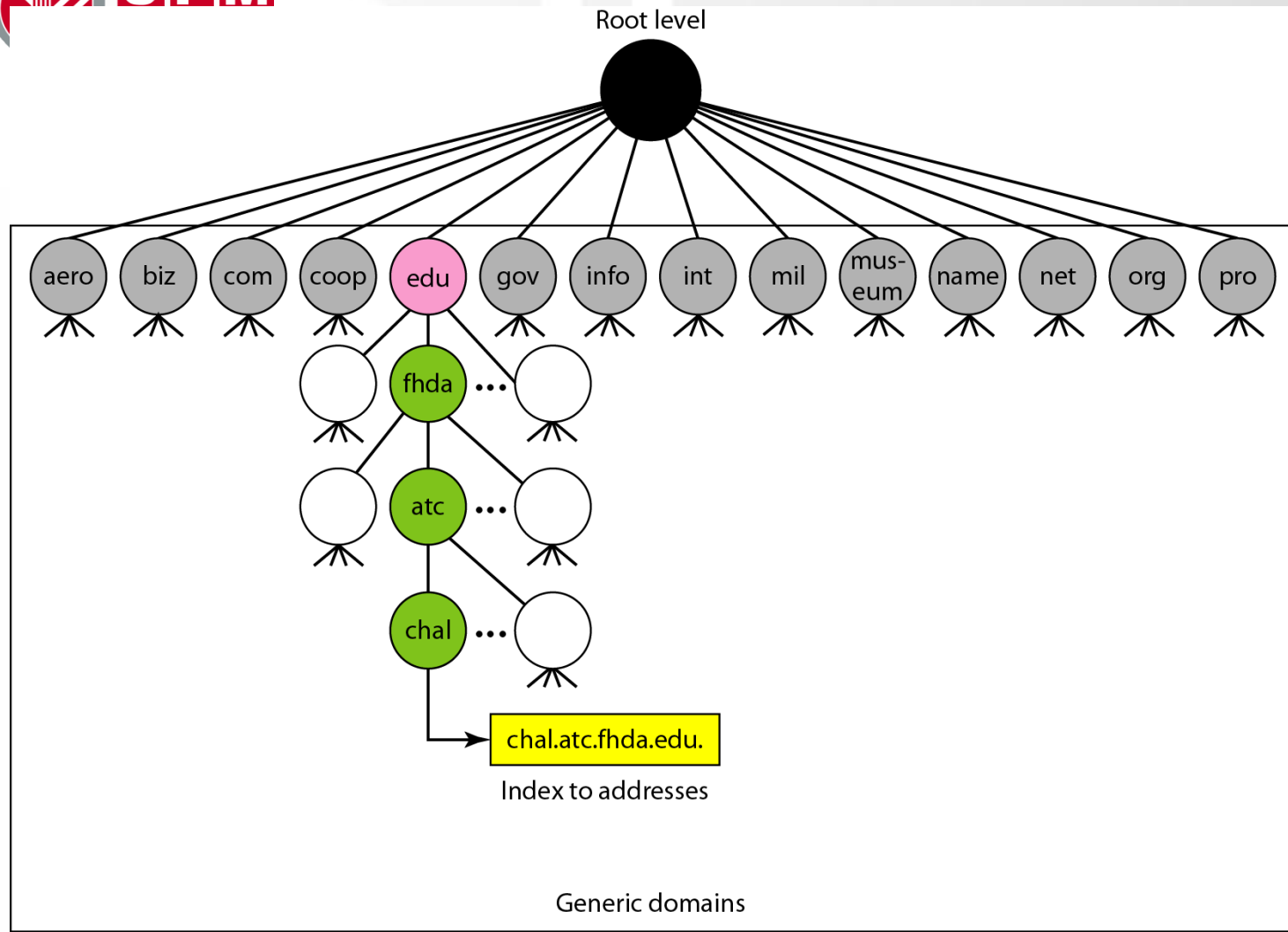


Table 25.1 *Generic domain labels*



<i>Label</i>	<i>Description</i>
aero	Airlines and aerospace companies
biz	Businesses or firms (similar to “com”)
com	Commercial organizations
coop	Cooperative business organizations
edu	Educational institutions
gov	Government institutions
info	Information service providers
int	International organizations
mil	Military groups
museum	Museums and other nonprofit organizations
name	Personal names (individuals)
net	Network support centers
org	Nonprofit organizations
pro	Professional individual organizations



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Figure 25.10 *Country domains*

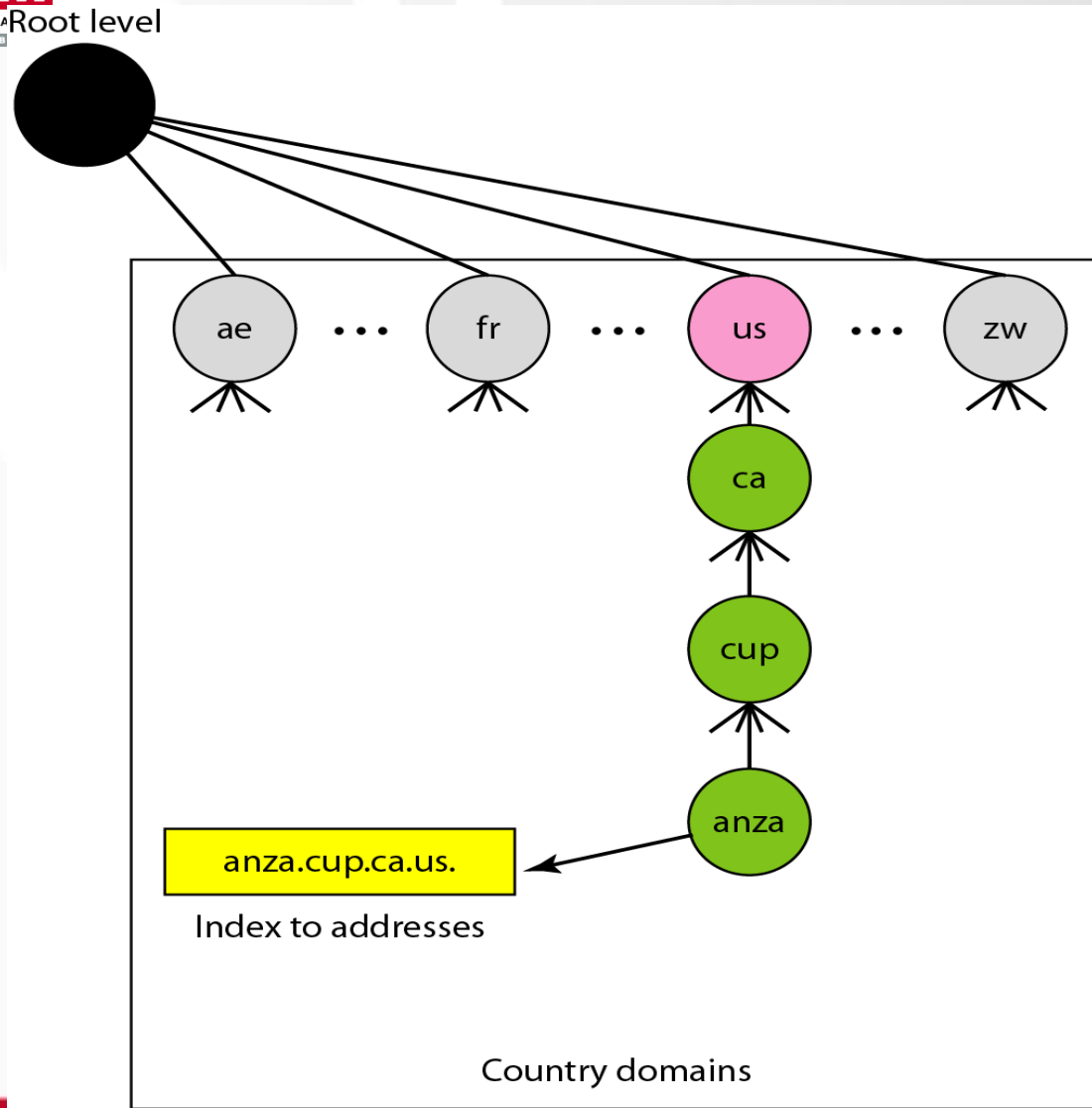


Figure 25.11 *Inverse domain*

